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Topical Review

Advances in nonlinear photonic devices based on lithium niobate waveguides

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Abstract

The nonlinear optical waveguide is one of the essential components of modern photonic integrated circuits. A high-quality lithium niobate (LN) waveguide has recently become available due to advances in thin-film LN materials and the associated fabrication techniques. In the past few years, LN waveguide-based nonlinear photonic devices have been intensively investigated due to their ultra-low loss and large index-contrast features. Here we review the recent progress in LN waveguide-based nonlinear photonics devices, including both passive and active components. We believe that LN-based nonlinear photonic devices will make a profound impact on modern photonic society.

Keywords: waveguide, fabrication techniques, photonic integrated circuits

(Some figures may appear in colour only in the online journal)

1. Introduction

There is a relentless demand for higher information exchange and processing accompanied by the rapid development of information technology. Photonic integration dramatically improves the performance of current optical systems, by reducing size, power consumption, cost, and manufacturing complexity at chip-scale. In the past decades, photonic integrated circuits (PICs) have developed remarkably and been successfully applied in many optical applications [1]. In particular, the optical waveguide has become one of the essential component of PICs [2]. As a building block of PICs, an optical waveguide can confine and direct photons in the desired directions at light speed, via the physical principle of total internal reflection. It is well known that the optical waveguide can transport information carried by photons, and has been demonstrated to be stable, and robust to noise with ultra-high bandwidth.

The concept of using an optical waveguide as the fundamental structure to construct PICs dates back to the late 1960s [3–5]. After the invention of the laser, optical waveguides became an elementary component for long-distance light transmission [6]. Since then, investigations on fundamental theory, materials, fabrication techniques, and functional devices of optical waveguides (such as couplers, interferometers, resonators, and so on) have been thoroughly and widely performed. Specifically, to promote the functionalities and associated performances of PICs, various materials have been proposed, including silicon [7–9], compound semiconductors (AlN, GaAs, etc) [10, 11], lithium niobate (LN) [12–16], Si_3N_4 [17–19], and polymer [20]. Generally, optical waveguides can be either passive (such as: directional coupler,

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Table 1. List of key properties of LN compared to some other commonly used materials in photonics.

Material	Point group symmetry	Optical index @1550 nm	Absorption coefficient @775 nm	Second-order nonlinear coefficient (pm/V)	Nonlinear index of refraction (m^2/W)	EO coefficient (pm/V)
$LiNbO_3$ [12]	3m	2.21(o)[a] 2.13(e)[b]	0	$d_{31} = -4.3$ $d_{33} = -27$ $d_{22} = 2.1$ @1064 nm	1.8×10^{-19} @1550 nm	$r_{13} = 9.6$ $r_{22} = 6.8$ $r_{33} = 30.9$ $r_{51} = 32.6$
Si [7]	m3m	3.49	0.008	0	5×10^{-18} @1550 nm	0
Si_3N_4 [17]	Amorphous	2.00	0	0	2.5×10^{-19} @1550 nm	0
AlN [11]	6mm	2.12(o) 2.16(e)	0	$d_{31} = 1.6$ $d_{33} = 4.7$ @1550 nm	2.3×10^{-19} @1550 nm	$r_{13} = 0.67$ $r_{33} = -0.59$
GaAs [10]	−43m	3.37	0.091	$d_{36} = 170$ @1064 nm	2.6×10^{-17} @1550 nm	$r_{41} = 1.43$

[a] ordinary wave.
[b] extraordinary wave

interferometer, modulator, resonator, switch, and multiplexer) or active (such as amplifier and laser).

Traditionally, optical waveguides can be fabricated through proton exchange, ion implantation, and metal diffusion [21–23]. However, these fabricated waveguides usually have small refractive index- contrast between the waveguide core and cladding, leading to poor confinement of guided photons. This point has restricted the shrinking of waveguide size down to wavelength scale and prevented the functionalities of optical waveguides from further progressing, thereby becoming the obstacle to massive photonic integration. Therefore, the fabrication of optical waveguides with large refractive index contrast, enabling strong confinement of photons and enhanced light-matter interactions, is highly desirable. Fortunately, benefiting from the advancements of materials and fabrication techniques, optical waveguides can be fabricated via various chemical/physical fabrication techniques.

In particular, nonlinear optical functionalities are crucial for constructing waveguide-based nonlinear photonic devices. When light travels in a nonlinear waveguide with a considerable power level, various nonlinear effects, including second-order effects (second-harmonic generation (SHG), sum-frequency generation (SFG), difference-frequency generation (DFG)), third-order effects (cross-phase modulation (XPM), self-phase modulation (SPM), four-wave mixing (FWM)), and electro-optic (EO) modulation, will take place. Nonlinear optical waveguides have been fabricated in various nonlinear optical materials, such as SiO_2, Si, Si_3N_4, InP, TiO_2, AlN, and LN. To date, nonlinear optical waveguides have been employed in various applications, including frequency converters, high-speed optical switches, high-speed modulators, waveguide amplifiers in telecommunications, and quantum light sources [24–28].

Specifically, LN material has both large refractive index and large nonlinear optical coefficients, which are promising for both strong confinement of photons and dramatically enhanced nonlinear optical interactions (shown in table 1). Compared with Si, LN material has larger EO coefficients and lower absorption losses in the visible and mid-infrared range [29], enabling various optical applications in the near-infrared range. It should be noted that high-quality thin-film lithium niobate on insulator (LNOI), prepared from ion slicing and wafer bonding, has recently emerged as an ideal candidate material, due to its excellent optical and EO properties [12–16]. The thin-film LN platform enables the fabrication of ultra-low loss, high index-contrast, nanophotonic, nonlinear LN waveguides [30–39] which opens a new avenue for high-performance on-chip devices and will make a profound impact on modern photonic society.

Benefitting from the development of thin-film LN platforms and continuous progress in nanofabrication techniques, nonlinear LN optical waveguides with record-low optical loss and large refractive index-contrast have been successfully fabricated. In previous work, an optical ridge LNOI waveguide was reported by Hu *et al* [13] in 2012 with a propagation loss of 6.3 dB cm^{-1}. And the latest report shows that optical loss of single-mode LNOI waveguides could be as low as 0.029 dB cm^{-1} [40], while this value could be 0.027 dB cm^{-1} for multimode waveguides [31]. Besides, such a low-loss LN waveguide enables extensive variety of novel EO and nonlinear optical devices including record-fast EO modulators [41–46], broadband frequency comb sources [47–49], integrated spectrometers [50], and highly efficient wavelength converters [51–54], as well as supercontinuum light sources [55–57].

LN waveguide devices are now still in the early stage of rapid development, which attracts broadband interest from both academic society and industry for better performances with multi-functional PICs [12–16, 58–63]. In this review, we briefly introduce the properties of the LN waveguide (section 2). Then we concisely describe and compare different fabrication techniques (section 3). Next, we introduce recent development of on-chip waveguide devices in various applications (section 4). Lastly, we summarize and consider the prospects of this research field.

2. Waveguide properties

An ideal LN waveguide would be of rectangular shaped (figure 1(a)). Such morphology should provide strong confinements for both transverse electric (TE) and transverse magnetic (TM) polarized modes. The traditional LN waveguide (figure 1(b)) suffers from indistinct boundary and small refractive index-contrast between waveguide core and cladding, leading to large mode area. With the advancement of fabrication techniques, a LN ridge waveguide, as shown in figure 1(c) with its morphology close to an ideal one, has been realized for constructing various integrated photonic devices. Various numerical methods can be used to investigate the waveguide properties, such as mode profile and dispersions. And the waveguide properties can readily be modified via waveguide morphology engineering. Figure 2 shows the calculated electric field intensity distribution of a LN ridge waveguide with 1 μm top width, 600 nm height, and $\sim 80^\circ$ sidewall angle. Both fundamental and high-order TE and TM modes are supported inside such a waveguide [63]. The calculated effective mode area of the fundamental TE and TM modes is extremely small, which is promising for strong light–matter interactions. As the transverse dimension of the waveguide shrinks, it becomes single-mode operation for only supporting fundamental modes.

It is crucial to manipulate the dispersion of the LN waveguide via morphology engineering. In the context of nonlinear optical processes, it is vital to realize phase matching conditions (or equivalently, momentum conservation) for the associated conversion efficiencies. Besides, manipulations of group velocity (GV) and group velocity dispersion (GVD) are essential for many nonlinear processes. For example, anomalous GVDs are usually required for maintaining temporal pulse shape so as to ensure sustained nonlinear optical conversions, such as supercontinuum generation (SCG) [64] and optical frequency combs (OFCs) [49]. Additionally, broadband SHG can be achieved when GVD is close to zero between the fundamental and second-harmonic (SH) wavelengths [65].

3. Fabrication technologies

As mentioned, various waveguide-based integrated nonlinear optical devices have been successfully fabricated on the platform of LNOI. This section will introduce several fabrication techniques for waveguide structures, including wet etching, dry etching, mechanical fabrication methods, rib-loading methods, and domain eversing methods. High-profile LN waveguides have been fabricated via the above techniques with record-low propagation losses (see table 2).

3.1. Wet etching method

The wet etching method was among the early attempts at fabrication of photonic microstructures of nonlinear materials, such as lithium tantalate ($LiTaO_3$), potassium titania phosphate, and potassium niobate ($KNbO_3$). The procedure of wet etching can be summarized into three main steps (figure 3). Firstly, a mask, usually made of chromium, is deposited on the top of LN and then helps form a waveguide structure by lithography. Secondly, liquid chemicals, for example, hydrofluoric acid (HF) or potassium hydroxide (KOH), are used to etch LN. Thirdly, the covering mask is removed to get the remaining LN ridge waveguide. Via this technique, Hu *et al* fabricated a ridge waveguide with a height of up to 8 μm and a width between 4.5 and 7.0 μm: associated propagation losses in a 6.5 μm wide and 8 μm high structure are typically 0.3 dB cm^{-1} for TE and 0.9 dB cm^{-1} for TM polarization at telecommunication wavelength (1.55 μm) [66]. Wang *et al* demonstrated a ridge waveguide with a width between 25 and 45 μm and a height of 1.2 μm by using MeV O^+ and Si^+ ion implantation with a wet etch technique [68]. Additionally, Xiang *et al* reported a 10 mm long LN ridge waveguide fabricated by MeV copper ion implantation and ion beam-enhanced wet etching, in which its SCG was with the broadest coherence length of 5.2 μm at around 800 nm, spanning from 693 to 995 nm [69].

Wet etching was demonstrated with some advantages, including high selectivity and low cost. It should be noted that the wet etching method will produce LN waveguides with tilted sidewalls. This is due to the anisotropic etching processes. For the uniaxial crystal structure of ferroelectric LN, the etching rate of HF acid on the positive electric surface (perpendicular to the *c* axis) is slower than that on the negative electric surface [70]. It is difficult to fabricate a LN waveguide with a width below 1 μm owing to such anisotropic features of wet etching [66]. As the wet etching rate of LN is quite slow ($\sim$3 nm h^{-1}) [66], several techniques, including Ar ion implantation, proton exchange, and ion diffusion, are utilized for pretreating LN so as to increase the etching rate [71–73]. For example, Ar ion implantation used before wet etching by KOH could amorphize the LN material and boost the etching rate [42]. To overcome the imperfection of the wet etching technique, researchers turned to the dry etching method as discussed below.

3.2. Dry etching method

Dry etching methods attract increasing attention these days and are becoming one of the most promising methods for fabricating high-quality LN waveguides. Dry etching is highly controllable and anisotropic. Initially, there was not much progress in dry etching due to the chemical stability of LN bulk crystal [74, 75] in earlier days. Along with the rapid developments in the LNOI platform, an ultra-low loss

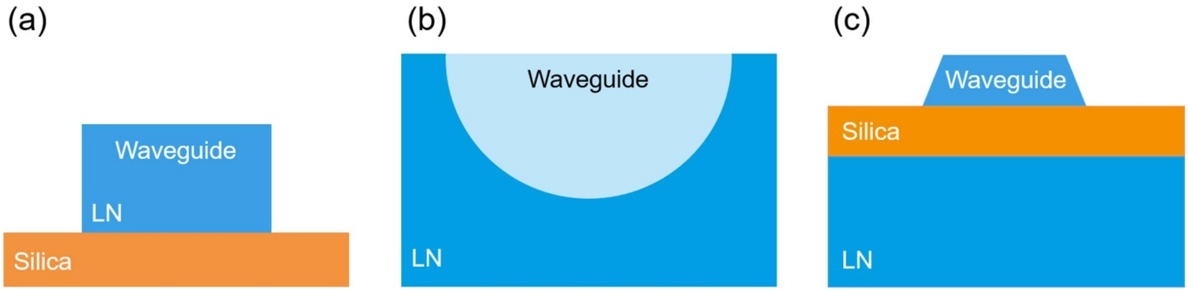

Figure 1. Schematic of LN waveguides with (a) ideal rectangular, (b) traditional diffused, and (c) ridge cross-sections.

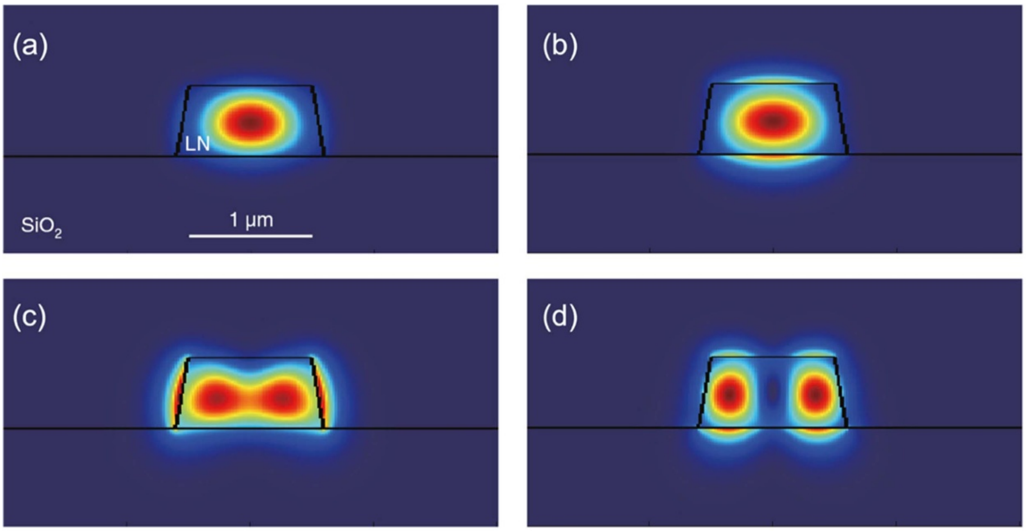

Figure 2. Calculated electric field intensity distributions of a LN ridge waveguide with 1 μm top width, 600 nm height, and ~80° sidewall angle. (a) Fundamental TE mode; (b) fundamental TM mode; (c) second-order TE mode; (d) second-order TM mode. The black lines indicate the profile of the waveguide and substrate (Reproduced from [63]. CC BY 4.0).

Table 2. Propagation losses of ridge waveguides with different fabrication techniques.

Year	Fabrication technique	Loss (dB/cm)	Wavelength (nm)	References
2007	Wet etching	0.3 TE	1550	LNOI [66]
2018	RIE	0.41 TE	1550	LNOI [15]
2018	Ar ion milling	0.268 TE	1550	LNOI [38]
2018	CMP[a]	0.04 TE	1500	LNOI [32]
2020	Lithography-assisted CMP	0.029	1550	LNOI [40]
2011	Dicing	0.2	1550	PPLN[b] [67]

[a] Chemo-mechanical polishing.
[b] Periodically polarized lithium niobate.

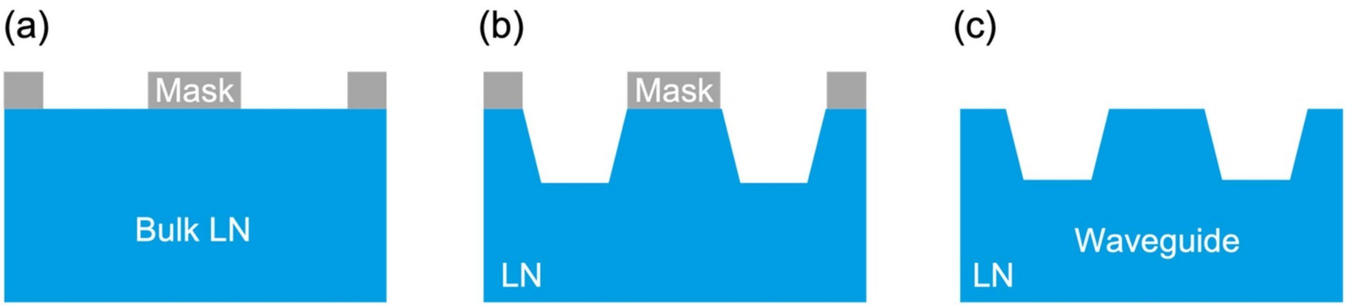

Figure 3. A typical wet etching process based on bulk LN. (a) Deposit a patterned mask on bulk LN. (b) Etch LN by liquid chemicals. (c) Remove the mask and a ridge LN waveguide structure is formed.

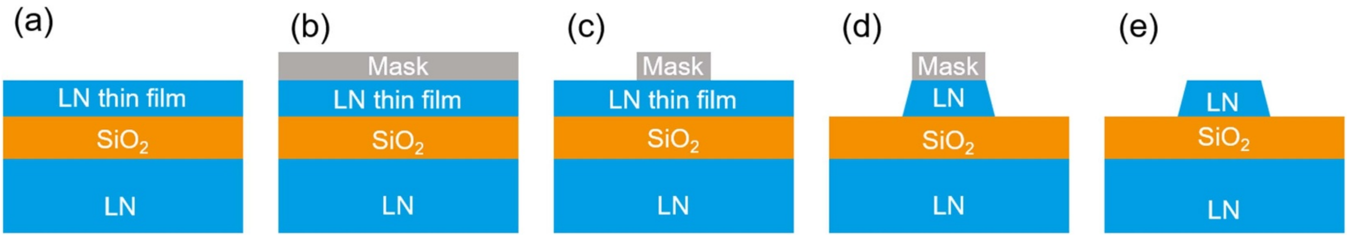

Figure 4. A typical dry etching process based on LNOI. (a) Prepare a clean LNOI wafer. (b) Deposit an etching mask on LNOI. (c) Pattern the etching mask by EBL. (d) Form the waveguide geometry by RIE or inductively coupled plasma (ICP)-RIE methods. (e) Remove the mask.

waveguide with submicrometer dimensions based on LNOI has been successfully achieved by the latest dry etching technique [15].

The dry etching method can be summarized into four basic steps, as shown in figure 4. Firstly, resisting materials, such as chromium [75], amorphous silicon [42], and others [15, 76], are deposited as an etching mask on LNOI. The use of different materials as etching masks corresponds to different etching processes. Secondly, photonic patterns are transferred to the masks by photolithography or electron-beam lithography (EBL). In some cases, an annealing process is applied in order to improve the sensitivity of etching orientation or reduce the sidewall's roughness [75]. Thirdly, etching of LN materials, via processes including reactive ion etching (RIE) [77–83], Ar ion milling [38, 84], and focused ion beam (FIB) milling [13, 85], is conducted to fabricate the optical waveguide structure. Finally, the optical waveguide is obtained after removing the covering mask.

To be noted, RIE and Ar ion milling are complementary metal oxide semiconductor (CMOS)-compatible nanofabrication techniques and both have been demonstrated to be efficient for dry etching of LNOI. RIE intrinsically suffers from redeposition of LiF on the LN surface during the etching process and contributes to rough surfaces in the waveguide and the associated scattering losses. To reduce the redeposition of LiF and improve the surface smoothness, Ar gas is added to bombard the LN material and enhance the sputtering rate during the etching process [77, 81, 83]. In addition, inductively coupled plasma RIE (ICP-RIE) is also used for etching LNOI to obtain the waveguide structure [86]. The ICP is utilized for deep etching by generating high-density plasma. Recently, Krasnokutska *et al* reported an LNOI single-mode waveguide with a propagation loss of 0.41 dB cm^{-1} at 1550 nm wavelength [15]. Pan *et al* achieved an LNOI microring cavity with a quality factor of 1.78×10^6 at 1572.6 nm wavelength in 2019 [86].

The Ar ion milling method has been employed to fabricate ridge waveguides on LNOI since 2005 [78]. Common photoresist is usually used as an etching mask in Ar ion milling [86]. To obtain a high-quality photoresist pattern with a smooth surface and sidewall, one should choose high-precision lithographic techniques, such as EBL or ultraviolet (UV) lithography. In 2018, Siew *et al* achieved an LNOI ridge waveguide with a propagation loss (TE mode) of 0.268 dB cm^{-1} around the 1550 nm wavelength by using EBL before Ar ion milling [38]. Ulliac *et al* utilized UV lithography to form a photomask before etching, and obtained single-mode propagation with optical losses of 5 dB cm^{-1} [84]. Notably, ICP-RIE is also frequently used for patterning the photoresist in the Ar ion milling method. Zhang *et al* used a commercial ICP-RIE tool to transfer the patterns into the thin-film LN and achieved a propagation loss of 0.027 dB cm^{-1} at 1590 nm wavelength [30]. In 2019, a similar etching process was developed by Desiatov *et al* to achieve a low propagation loss of 6 dB m^{-1} and microring resonator of intrinsic quality factor $\sim$11 million [34].

FIB milling, a pure physical etching process using ion particles, has also been applied to etch LN. FIB milling can achieve a relatively high resolution ($\sim$10 nm) and provide smoother surfaces than the Ar ion milling method. An LNOI ridge waveguide, fabricated through FIB milling by Guarino *et al* [13], showed a better surface than the one realized by Ar ion milling. However, FIB milling also suffers from redeposition resulting in a striking roughness of the top surface [60]. It is worth noting that one primary limitation of FIB milling is the low etching rate due to its scanning and exposure procedures. To overcome that drawback, coarse milling mode is used to reduce the redeposition and promote fabrication efficiency. Subsequently, a two-step FIB milling method, starting with rough milling and followed by fine milling, has been developed to obtain a smooth surface with shorter time span [85].

The feasibility of the dry etching method has been demonstrated but several deficiencies still remain unsolved. Researchers still work to reduce redeposition and improve both surface smoothness and fabrication efficiency. As the propagation loss of waveguide structures fabricated by ionic dry etching mainly comes from the surface roughness instead of material absorption, researchers might obtain high-quality waveguide structures through improved etching methods or by other fabrication techniques.

3.3. Mechanical fabrication method

Considering the chemical inertness of LN, mechanical fabrication methods have been used to ultimately reduce the roughness and associated scattering losses. Mechanical methods generally include chemo-mechanical polishing (CMP) [87, 88], diamond dicing [33, 67, 89–91], and ultra-precision cutting [92, 93]. CMP, a physical method used in the

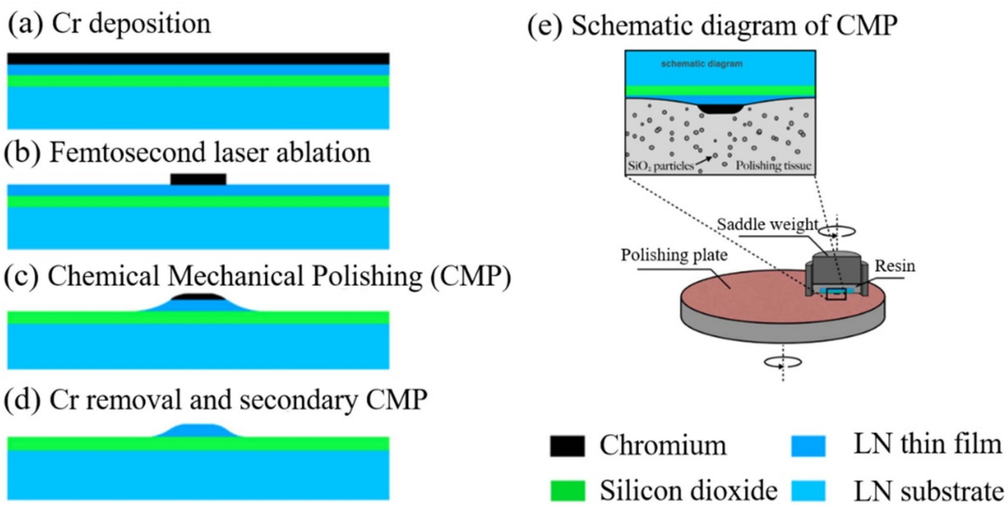

Figure 5. A typical CMP fabrication process for LN waveguide fabrication. (a) Deposit Cr thin film. (b) Pattern the Cr thin film using a femtosecond laser. (c) Form the waveguide by CMP. (d) Remove the Cr mask by wet etching and obtain a smoother surface by secondary CMP. (e) Schematics of typical CMP fabrication process (Reproduced from [31]. CC BY 4.0).

semiconductor industry to smooth the wafer surface, is also utilized in LNOI nanofabrication, and is expected to fabricate a surface as smooth as dry etching. Figure 5 shows the fabrication processes of the LNOI waveguide by CMP, which can be summarized into three main steps [31]. Firstly, a Cr layer is deposited on LN as a mask transferred by femtosecond laser or UV lithography. Secondly, a wafer polishing machine is used to remove the exposed LN and shape the waveguide. Thirdly, the Cr mask is removed by the Cr etching solution (Cr etchant, Alfa Aesar GmbH) and a secondary CMP stage is used in order to complete the waveguide fabrication. For CMP without using any ion beam etching, the surface roughness on the nearly vertical sidewall can be effectively avoided. Benefiting from the technique, the propagation loss of the waveguide can be kept as low as 0.04 dB cm^{-1} [32], while the quality factor of microring resonator can reach 3×10^6 [87]. By covering a Ta_2O_5 cladding, researchers have fabricated a single-mode LNOI ridge waveguide with a propagation loss of 0.029 dB cm^{-1} [40]. However, to be noted is that the aspect ratio (generally lower than 1.5) of the fabricated waveguide is restricted by metal mask [40].

In order to overcome the shortcomings of CMP, diamond dicing, a direct machining method with a high aspect ratio, is employed to fabricate LN waveguides. Figures 6(a)–(f) show the fabrication process. The shape of the cutting blade determines sidewall angle, top width of waveguides, and gap distance between two adjacent waveguides [90]. Courjal *et al* realized a high aspect ratio (>500) LNOI ridge waveguide of 2 μm top width and 500 μm depth with 0.5 dB cm^{-1} propagation loss [94]. Figures 6(g) and (h) shows a MgO:LN ridge waveguide with a width of 6 μm and depth of 30 μm possessing a propagation loss of 0.2 dB cm^{-1} [67]. However, diamond dicing cannot process complex structures (e.g. couplers, rings, crosses, and tapers) due to the defined moving direction of dicing blade. The method of ultra-precision cutting can overcome such a limitation and produce a curved waveguide. A typical work reported by Takigawa *et al* showed an LNOI ridge waveguide with a bending radius of 300 μm [92].

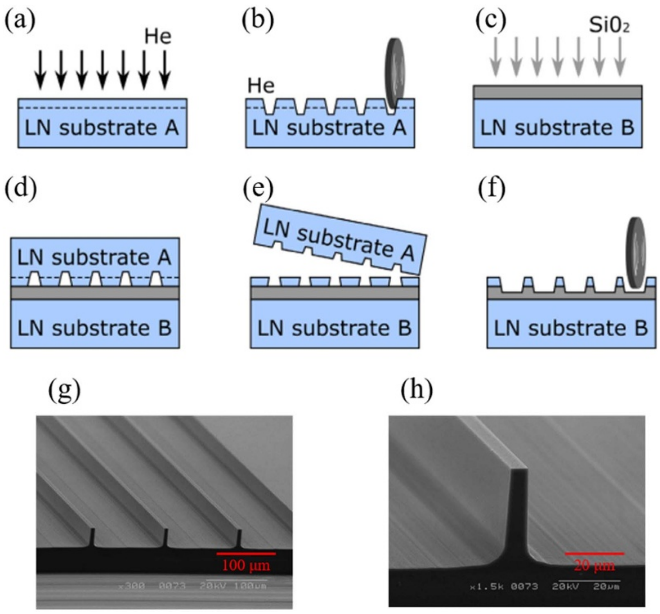

Figure 6. A diamond dicing fabrication process for waveguides. (a)–(f) Schematic diagram of diamond dicing (Reproduced from [90]. CC BY 4.0). (g) An SEM image of a ridge waveguide fabricated by diamond dicing and (h) magnified image of one ridge waveguide (Reprinted with permission from [67] © The Optical Society).

3.4. Rib-loading method

The methods mentioned above are direct fabrication on LN, which all require extremely precise control. Alternatively, a pragmatic method for fabricating waveguide structures is to bond a rib waveguide of some other material with LN, so

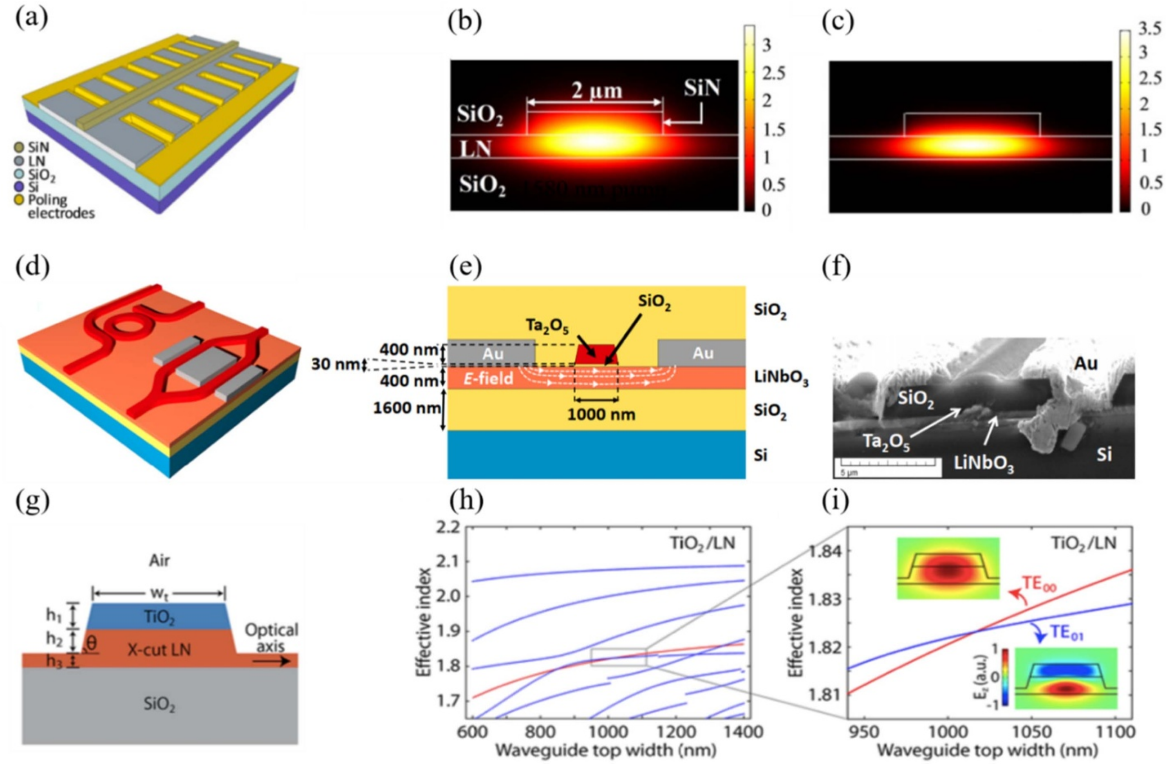

Figure 7. Rib-loading waveguide. (a) Schematic view of SiN rib based on LNOI. (b) The fundamental TE modes at 1580 nm and (c) the second-harmonic modes at 790 nm (Reproduced from [37]. CC BY 4.0). (d) Schematic view, (e) cross-section view, and SEM image of Ta_2O_5 rib waveguides (Reproduced from [95]. CC BY 4.0). (g) Schematic view of TiO_2/LN semi-nonlinear waveguide. (h) Dispersion curves of modes at 1550 nm and at 775 nm. (i) Phase matching points for SHG between TE_{00} at 1550 nm and TE_{01} at 775 nm ([100] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim).

as to form a hybrid waveguide structure. This method avoids direct etching of LN material, providing favorable feasibility by removing the fabricating limitations on LNOI. The introduction of top rib materials, such as Si, SiN, Ta_2O_5, TiO_2 and polymer, allows for use of mature fabrication methods, some of which are compatible with semiconductor fabrication techniques [95–99]. Figures 7(a)–(c) shows a periodically poled SiN rib-loaded waveguide based on LNOI by Rao *et al* in 2016 [37]. As can be seen, the mode fields could be strongly restored in the volume of the hybrid ridge waveguide, which is promising for enhanced light–matter interaction. Besides, Rabiei *et al* achieved a Mach–Zehnder modulator made of Ta_2O_5 rib waveguides based on LNOI, which showed a Y-junction integrated with Au electrodes, as shown in figures 7(d)–(f) [95]. Of note, a semi-nonlinear TiO_2:LN waveguide was proposed with large modal overlap integral and high-efficiency nonlinear frequency conversion, as shown in figures 7(g)–(i) [100]. In addition, Yu *et al* demonstrated the Mach–Zehnder interferometers (MZIs) and multimode directional couplers based on an ultra-low loss, etchless, polymer:LN-integrated photonic platform [101].

3.5. Domain reversing method

It is known that the spontaneous electric polarization of LN, as ferroelectric material, can be reversed by an external electric field that is stronger than its internal coercive field. By carefully modulating the microscopic ferroelectric polarization, one may readily achieve quasi-phase matching (QPM) for efficient nonlinear optical conversions. The fabricated structure is the so-called periodically poled lithium niobate (PPLN), which can compensate for the velocity mismatch between the interacting waves by periodically altering the ferroelectric direction of LN [104]. PPLN waveguides have been successfully applied to wavelength conversion [55, 105], optical signal processing [106], SCG [107], etc.

To fabricate a PPLN waveguide, one could directly change the ferroelectric domains of the LNOI waveguide by designing periodical microelectrodes fabricated on the top of a LN waveguide. The PPLN waveguide structure is formed after applying the proper voltage to these electrodes for a time. In addition, a PPLN waveguide can be fabricated with the assistance of a conductive atomic force microscope, as shown in figure 8(a) [102]. Alternative approaches are also available, including slicing PPLN thin film from a bulky PPLN LN crystal [110], direct laser writing [111], and the specific growth technique [112]. As shown in figure 8(b), Li *et al* achieved a single-crystal PPLN thin film via the slicing method [103], while PPLN rib-loading waveguides were successfully fabricated on the top of PPLN by methods mentioned above [51, 54, 113].

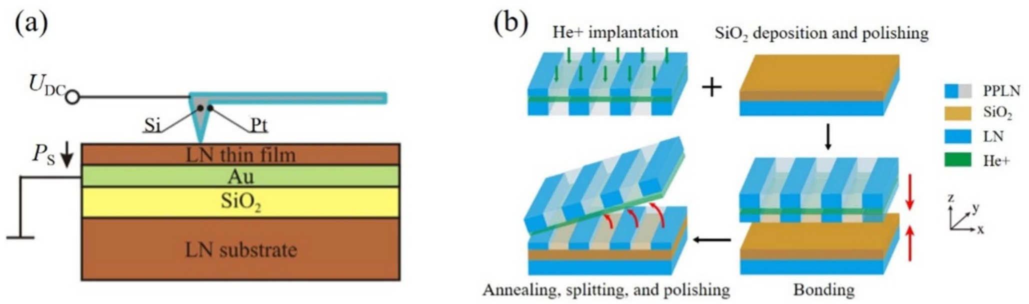

Figure 8. Domain reversing fabrication methods for PPLN. (a) Schematic diagram of local domain poling fabrication by AFM (Reprinted from [102], with the permission of AIP Publishing). (b) Schematic diagram of a slicing PPLN thin film bonding to the substrate (Reprinted with permission from [103] © The Optical Society).

4. Nonlinear optical devices

Benefitting from the large linear refractive index and nonlinearity, LN waveguides can provide strong confinements of photons and become an ideal platform for various nonlinear optical effects. In this section, various nonlinear optical devices based on waveguide structures are discussed, including nonlinear optical conversions, quantum light resources, SCG, EO devices, OFC, optical delay lines (ODLs), and so on.

4.1. SHG

LN has exhibited large second-order nonlinearity and a wide transparent window [24–28]. With the rapid development of fabrication techniques, second-order nonlinear optical conversions are investigated based on LNOI platforms, such as SHG, SFG, DFG, spontaneous parametric downconversion, and optical parametric oscillation [114].

As one of the most common nonlinear optical phenomena, SHG can double the optical frequency of waveguiding optical waves via the second-order nonlinearity. Aside from the high second-order nonlinearity of LN, the LN waveguide can provide strong confinement of photons, which is favored for enhanced SHG. However, achieving phase-matching conditions is still requied to increase the efficiency of SHG. To coordinate phase velocities of fundamental and second-harmonic waves, various strategies have been previously investigated, which can be categorized into birefringent phase matching (BPM) [115, 116], modal phase matching (MPM) [117–119], and QPM [36, 37, 51, 55, 120].

BPM is rarely used in LN waveguides due to its birefringent dispersion feature and prohibited usage of the largest second-order nonlinear susceptibility ($\chi^{(2)}_{ZZZ}$). Alternatively, one can readily realize phase matching of SHG utilizing the largest nonlinear susceptibility ($\chi^{(2)}_{ZZZ}$) via MPM. Generally, MPM can only be achieved between fundamental mode at pump wavelength and high-order mode at SH wavelength [108, 109], due to material and waveguide dispersions. For example, phase matching was realized between fundamental TE mode at pump wavelength and third-order TE mode at second harmonics for a LN waveguide, with a calculated normalized conversion efficiency ~60% W^{-1} cm^{-2} and measured normalized conversion efficiency ~41% W^{-1} cm^{-2} demonstrated experimentally (figures 9(a) and (b)) [108]. To further promote the conversion efficiency, a LN microring resonator of loaded quality factor ~10^5 was introduced and the associated conversion efficiency was experimentally determined to be ~1500% W^{-1} for SHG (figures 9(c) and (d)) [109].

Another way to achieve MPM is to design a semi-nonlinear waveguide via the rib-loading method as mentioned above. Various materials (e.g. polymer, TiO_2, silicon) were introduced on the top of a LN waveguide [101, 122, 123]. Given the asymmetric profile of the waveguide, new phase matching conditions become available with a larger nonlinear overlap integral between the interacting modes. For example, we have fulfilled the phase matching condition in a polymer-LN hybrid waveguide with calculated normalized conversion efficiency ~148% W^{-1} cm^{-2} [123]. In addition, a reversed-polarization double-layer LN waveguide was also proposed with a large modal overlap integral and high-efficiency nonlinear frequency conversion experimentally (~5540% W^{-1} cm^{-2}) [122].

Aside from MPM, the QPM method has been successfully demonstrated to achieve high-efficiency SHG. It is widely known that there is phase mismatch between fundamental modes of pump and second harmonics due to material and waveguide dispersion. Via QPM, momentum mismatch can be compensated by the reciprocal vector of the periodically poled structure. In early work, a normalized conversion efficiency of SHG was achieved up to 150% W^{-1} cm^{-2} at 1550 nm based on a buried PPLN waveguide [124]. Based on an Mg/MgO-doped PPLN ridge waveguide, blue and UV light can be efficiently generated via SHG with a conversion efficiency of 0.9% W^{-1} [125]. As shown in figure 10(a), Wang *et al* fabricated *X*-cut, Mg-doped PPLN ridge waveguides via standard lithography and demonstrated a normalized conversion efficiency of SHG ~2600% W^{-1} cm^{-2} at a 1500 nm pump [51]. Most recently, a PPLN waveguide has demonstrated efficient SHG of blue light (456.5 nm) upon 1.62 mW pump power, with a record conversion efficiency of 33 000% W^{-1} cm^{-2} [126]. Furthermore, SHG was generated in a PPLN waveguide fabricated by pairs of deposited microelectrodes with ultra-high normalized conversion efficiency up to 3061% $W^{-1}cm^{-2}$

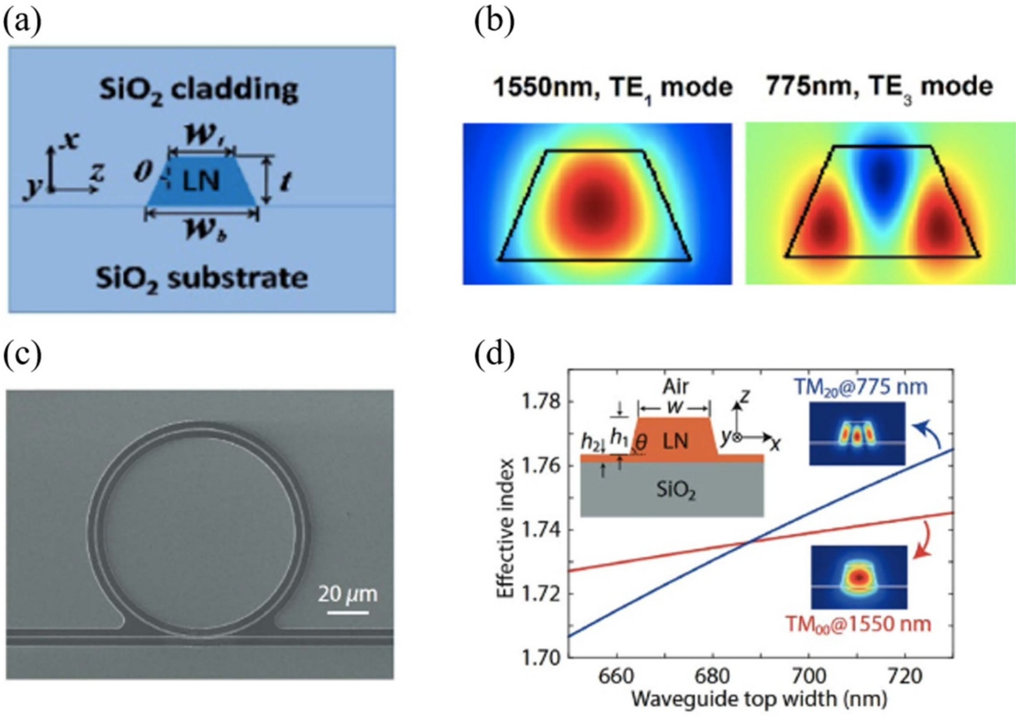

Figure 9. Modal phase matching between fundamental mode and high-order mode in LN waveguides. (a) Cross-section of LNOI waveguide. (b) Phase matching between TE_1 mode at 1550 nm and TE_3 mode at 775 nm (Reproduced from [108]. CC BY 4.0). (c) Scanning electron microscopy (SEM) image of LN microring. (d) Calculated modal effective indices of TM_{00} mode at 1550 nm and TM_{20} mode at 775 nm (Reprinted (figure) with permission from [109], Copyright (2019) by the American Physical Society).

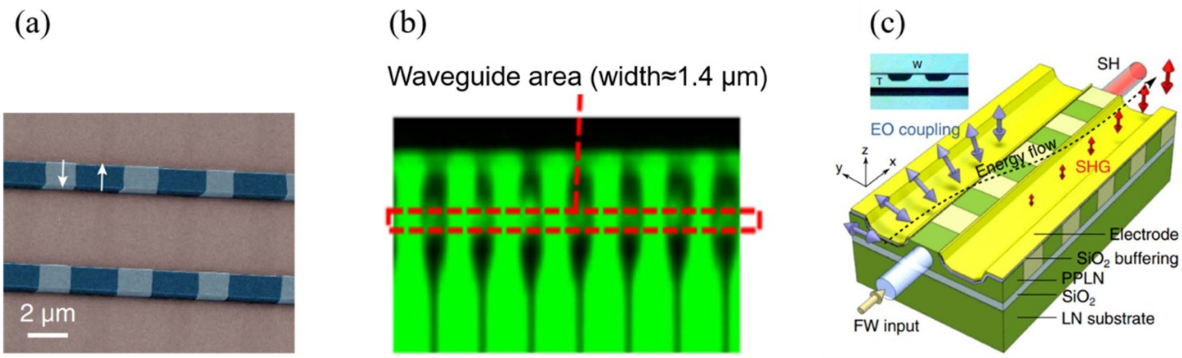

Figure 10. SHG devices based on PPLN ridge waveguides. (a) SEM images of PPLN ridge waveguides fabricated by electrode poling (Reproduced from [51]. CC BY 4.0). (b) The spatial evolution of the domain inversion with applied high-voltage pulses (Reprinted from [55], with the permission of AIP Publishing). (c) Schematic of SHG based on a PPLNOl ridge waveguide (Reprinted with permission from [121] © The Optical Society).

at 1470 nm (figure 10(b)) [55]. In addition, a PPLN ridge waveguide was fabricated together with cascading EO coupling (figure 10(c)), which successfully demonstrated integrated, electrically controllable, nonlinear devices with larger bandwidth [121]. Besides, Wang *et at* proposed a polarization-independent up-conversion protocol for single-photon detection at 1550 nm with a single thin-film PPLN waveguide, which achieved a normalized conversion efficiency up to 163.8% W^{-1} cm^{-2} [127]. To take advantage of both microring and QPM, microring resonators have been fabricated on PPLN with a quality factor of 1 million and associated conversion efficiency of 9×10^{-4} mW^{-1} (figures 11(a) and (b)) [113]. And high efficiency SHG was obtained in dual-resonant and periodically poled, *Z*-cut LN microrings, with a record-high conversion efficiency of SHG up to 250 000% W^{-1} cm^{-2} (figures 11(c) and (d)) [54].

Similar to PPLN waveguides, periodically nanostructured waveguides and associated devices have been demonstrated to fulfill the QPM condition [108]. By introducing periodic modulation of the waveguide width, periodically grooved LN waveguides were fabricated to achieve QPM between fundamental TE modes at both pump and second-harmonic

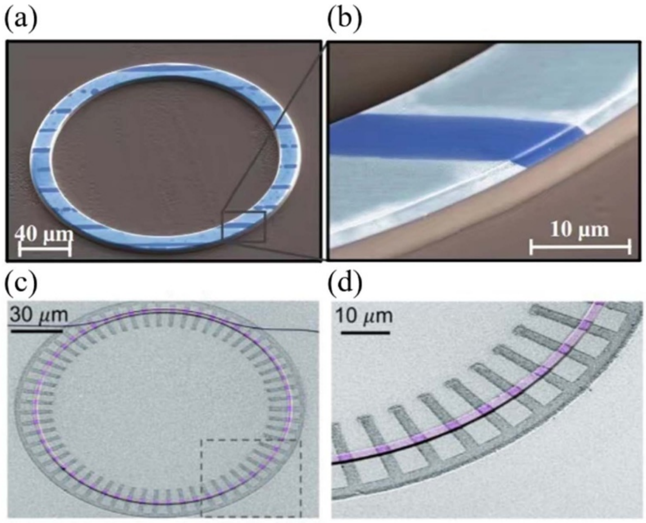

Figure 11. SHG devices based on PPLN ring waveguides. (a)–(d) SEM images of PPLN microring resonators and their zoom view (Reproduced from [54, 113]. CC BY 4.0).

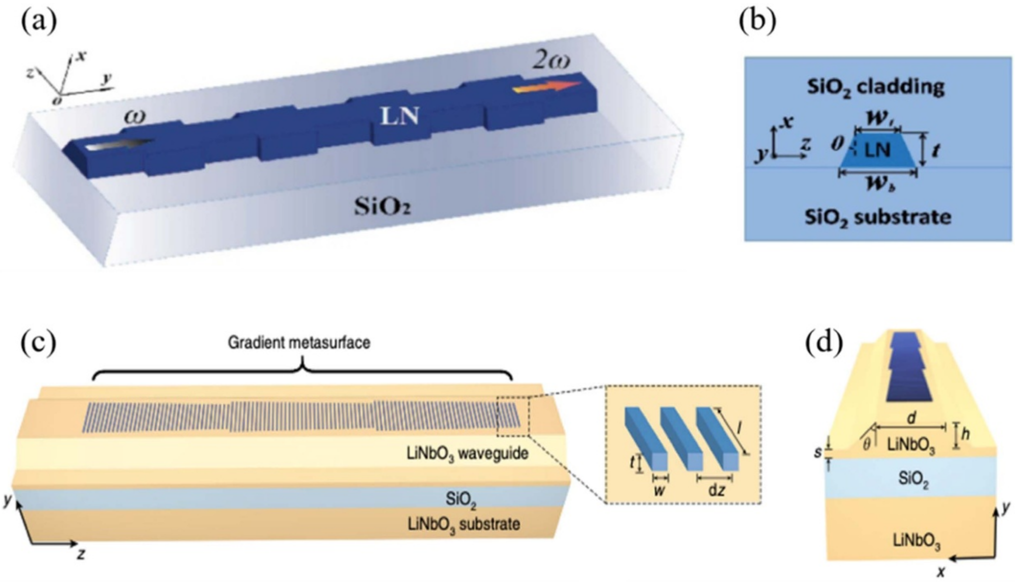

Figure 12. Periodically nanostructured LN waveguides. (a) 3D cartoon of the proposed periodically grooved LN waveguides (PGLN) structure. (b) Cross-section schematic of the *X*-cut LNOI waveguide (Reproduced from [108]. CC BY 4.0). (c) Schematic of LN waveguide with a gradient metasurface and (d) corresponding cross-section of the device (Reproduced from [116]. CC BY 4.0).

wavelengths, giving rise to a considerable conversion efficiency of $\sim$41% W^{-1} cm^{-2} [108] (figures 12(a) and (b)). The metasurface functioned as a converter of fundamental TE mode to high-order mode at second harmonics, thereby preventing the SHG from converting back to the pump wavelength. This method proved to be efficient even in the absence of phase matching with a normalized conversion efficiency of $\sim$1660% W^{-1} cm^{-2}, as shown in figures 12(c) and (d) [116].

4.2. SCG

SCG is a nonlinear optical process where photons are converted to broadband spectra. This process is usually generated through several nonlinear processes, including SPM, XPM, FWM, SHG, and dispersive wave generation [128]. It can be readily achieved by launching an ultra-short laser pulse into a dispersion-engineered nonlinear optical waveguide. The nonlinear optical waveguide is usually configured in the anomalous dispersion regime so as to ensure both spectral broadening and temporal compression. Surely, the LN waveguide is ideal for SCG due to its large material nonlinearity, i.e. $\chi^{(2)}$ ($d_{33} = 3 \times 10^{-11}$ m V^{-1}) and $\chi^{(3)}$ (1.6×10^{-21} m^2 V^{-2}), as well as tight confinement of photons. By careful design of waveguide geometry, 1.5-octave-spanning SCG and simultaneous SHG were demonstrated in LN waveguides upon 1560 nm femtosecond laser excitation (figure 13(a)) [56]. Furthermore, the QPM technique for the LN waveguide was

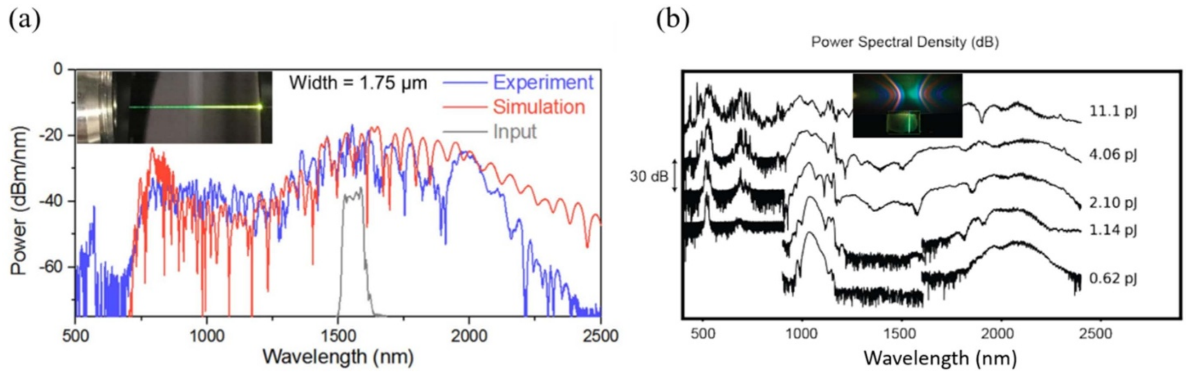

Figure 13. SCG in LN waveguides. (a) A SCG spanning over 1.5 octaves based on LNOI ridge waveguides (Reprinted with permission from [56] © The Optical Society). (b) SCG spectra at different pump pulse energies in the PPLN waveguides and inset image shows the photograph of SCG (Reproduced from [65]. CC BY 4.0).

utilized for efficient SCG (multi-octave spanning from 400 to 2400 nm) by reducing the pulse requirements from picojoules to femtojoules (figure 13(b)) [65].

4.3. EO effect

LN is an attractive material as a result of its excellent EO property. The large linear EO effect of LN (e.g. $r_{33} \sim 31$ pm V^{-1}), known as Pockels effect, can produce significant change of refractive index and associated phase shift of propagated photons upon application of external voltages, which enables optical modulation, sideband generation, and frequency shifting [58, 130]. Some existing approaches to realize gigahertz-scale frequency shifts are either limited to low efficiencies or frequencies, or are bulky, such as the acoustic-optics method and the all-optical wave mixing method. Benefitting from the thin-film LN waveguide geometry, microwave electrodes can be placed very close to the tightly confined photons, giving rise to fast-speed modulation with larger bandwidth and lower half-wave voltages, which can overcome the bottleneck of conventional bulk LN-based devices. Thus, EO frequency shifters have been realized that are controlled by using continuous and single-tone microwaves. To date, EO modulators, EO optical switches, and ODLs have been realized based on LN waveguides with outstanding performance.

The EO modulator, as one of the essential components in modern telecommunication networks, can translate electric signals into flying optical bits. Remarkable progress in increasing the bandwidth of the thin-film LN platform has been realized recently [35, 43, 131–138]. Typically, EO modulators can be realized based on structures such as the MZI, cascaded MZI, and Michelson interferometer modulator (MIM) [58]. An integrated MZI-type EO modulator has been demonstrated with half-wave voltage of 1.4 V and bandwidth of 45 GHz [42] and hard-decision forward error correction of 220 Gbit s^{-1} [139]. An integrated thin-film LN in-phase/quadrature EO modulator exhibited excellent performance (half-wave voltage ∼1.9 V, bandwidth ∼48 GHz, and optical loss ∼1.8 dB) and supported a modulation data rate up to 320 Gbit s^{-1} [35]. Based on an intensity-modulated LN modulator, researchers have realized transmission over 10.2 km single mode- fiber (SMF) at a record 700.4 Gb s^{-1} line rate and a record 538.8 Gb s^{-1} net rate with entropy-loaded probabilistically shaped (PS)-256-quadrature amplitude modulation (QAM) signals, and 200-GBaud PS-pulse-amplitude modulation (PAM)-16 signals with 650 Gb s^{-1} line rate and 484.5 Gb s^{-1} net rate [140]. A net data rate of 1.58 Tb s^{-1} on a single waveguide was demonstrated via a 102 GHz digital back-propagation (DBP) digital-to-analog converter (DAC) and a 100 GHz I/Q modulator with polarization-division multiplexed (PDM) 200 GBaud PS-64-QAM signals up to 7.92 b/s/Hz [141]. Most recently, a dual-polarization thin-film LN modulator enabled a record single-wavelength 1.96 Tb s^{-1} net data rate with a sub-1 V driving voltage, 110 GHz bandwidth, and ultra-low power consumption (1.04 fJ bit^{-1}) [142].

Notably, EO modulators with resonant structures have also been demonstrated with improved performance, even at the cost of reduced EO bandwidth [41] [109, 143–146]. Besides, in order to obtain a higher extinction ratio than a single MZI modulator, a cascaded MZI modulator was also proposed with an extinction ratio of 28 dB (figures 14(a) and (b)) [129]. To further increase the modulation efficiency and downsize the devices, a MIM was proposed to double the interaction length via refractive mirrors on parallel arms [129].

To take the advantages of both LN and Si (or other materials, such as SiN_x, Ta_2O_5, and chalcogenide glasses), hybrid EO modulators have been investigated, which can combine the scalability of other materials and outstanding EO feature of LN. The hybrid modulators can be categorized into two types, one is to bond thin-film LN on pre-patterned silicon photonic circuits and the other is to adopt a rib-loaded thin-film LN modulator. For example, He *et al* [43] demonstrated a hybrid MZI modulator consisting of two integrated Si–LN ridge waveguides and grating couplers, which showed an insert loss of 2.5 dB and a 3 dB EO bandwidth of up to 70 GHz, as shown in figures 14(c) and (d). Besides, Peter *et al* reported a LN/SOI hybrid modulator with 3 dB bandwidth exceeding 100 GHz via bonding a thin-film LN modulator on a silicon photonic platform, as shown in figure 14(e) [44]. Other high-performance LN on Si compact modulators have been demonstrated for optical interconnect and radio

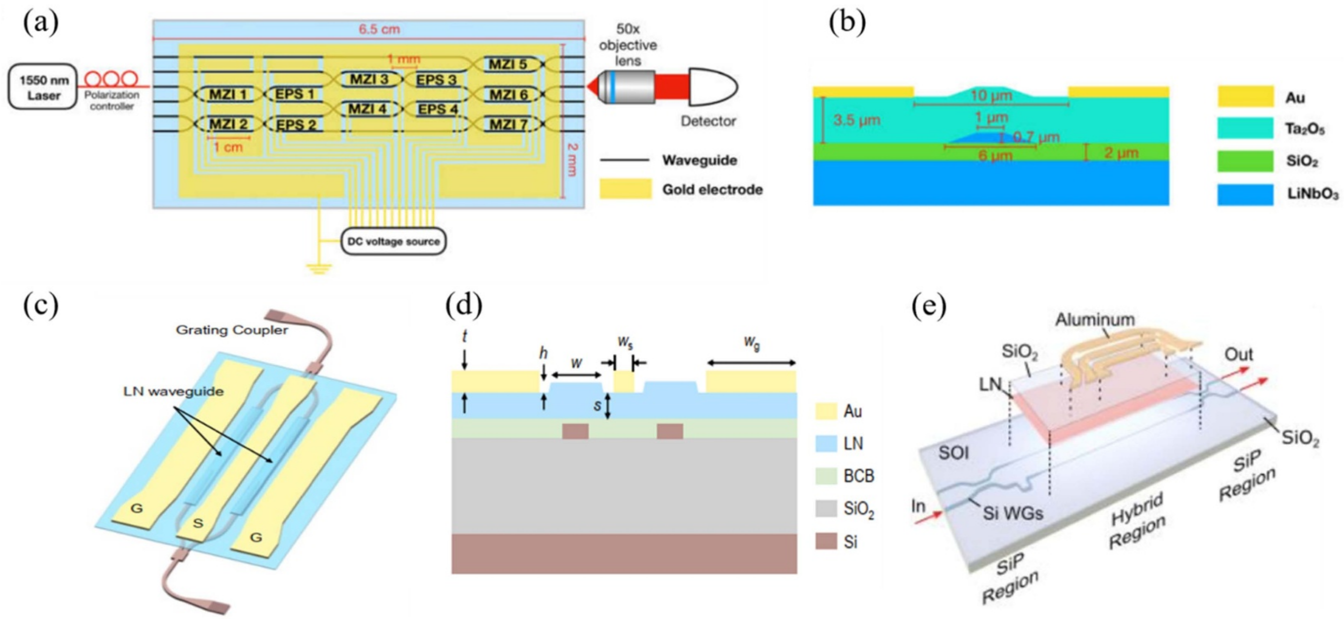

Figure 14. Typical EO modulators. (a) Schematic of a large-scale beam splitter consisting of cascading MZI EO modulators and (b) the cross-section of a LNOI waveguide (Reprinted with permission from [129] © The Optical Society). (c) Schematic of a hybrid Si/LN MZI EO modulator and (d) the cross-section view of the hybrid waveguide (Reproduced from [43]. CC BY 4.0). (e) Un-etched LN thin film was bonded to a Mach–Zehnder interferometer fabricated in Si and formed a LN/silicon on insulator (SOI) hybrid modulator (Reproduced from [44]. CC BY 4.0).

frequency applications. For example, Andrew *et al* reported an index-matched travelling-wave crystal ion sliced LN modulator with a measured EO response up to 500 GHz which makes it an ideal candidate for emerging technologies in the THz regime [45]. Similarly, Rao *et al* demonstrated a LN/Si hybrid compact modulator up to 50 GHz with lower drive voltages, small device footprints, and potential compatibility with CMOS [46].

4.4. OFC

An OFC is a kind of light source composed of equidistant frequencies, appearing as a temporal sequence of ultra-short pulses. As one striking breakthrough of laser technology, OFCs are adopted in various applications, such as optical clocks, precision spectroscopy, and coherent optical communications. Aside from OFCs based on mode-locked lasers, on-chip microcavity OFCs have emerged as a new type of OFC, benefiting from the rapid development of the high-quality microcavity [149]. And the thin-film LN platform provides a new opportunity for on-chip OFCs as a result of its large nonlinearities and strong confinements of photons [47, 49, 150].

Generally, the on-chip OFC originated from Kerr nonlinearity (also called the Kerr OFC), making use of the large third-order nonlinearity of LN through FWM. Benefitting from the high-quality microresonator of the thin-film LN platform, comb teeth could be efficiently generated at cavity mode frequencies. In an ultra-low loss *X*-cut LN waveguide with anomalous dispersion, a Kerr OFC spanning 700 nm was obtained [49]. To exclude the Raman scattering effect that could prevent soliton mode-locking, optical waveguides using the TE mode of *Z*-cut LN could readily achieve stable a OFC based on high-Q resonators spanning $>$200 nm at 1.5 μm (figure 15(a)) [147] and $>$300 nm at 2 μm (figure 15(b)) [148]. The Kerr OFC was also realized in PPLN ring resonators [150].

Another type of OFC takes advantage of the EO effect via the large second-order nonlinearity of LN. The EO OFC can be categorized into two: one is non-resonant and the other is resonant. By integrating an EO modulator on an *X*-cut thin-film LN platform with a small on-chip loss ($\sim$1 dB/channel), an OFC with large spectral range and 30 GHz spacing was obtained [151]. To overcome the restriction of the limited microwave power, a designed cascaded MZI would produce an OFC with spectral flatness of 0.89 dB though for a limited number of comb lines, with an on-chip optical loss less than 1.19 dB [152]. A resonant OFC is based on an optical cavity with an EO modulator. As photons circulate inside the cavity, they experience continuous modulating with expanded frequency combs. Such a process can be significantly enhanced when the frequency of microwave modulation matches the cavity mode spacing. An excellent low-loss LN microring resonant OFC with loaded $Q \sim 1.5$ million, consisting of 900 lines from 1560 nm to 1640 nm, was realized by launching continuous-wave laser light into a high-quality optical racetrack cavity (figure 15(c)) [47].

4.5. ODL and on-chip spectrometer

An ODL is an essential photonic component to generating various time delays in optical signals. It is highly desirable to develop ODLs with low propagation loss, strong confinement, large time delay, and EO tunability on the microchip scale.

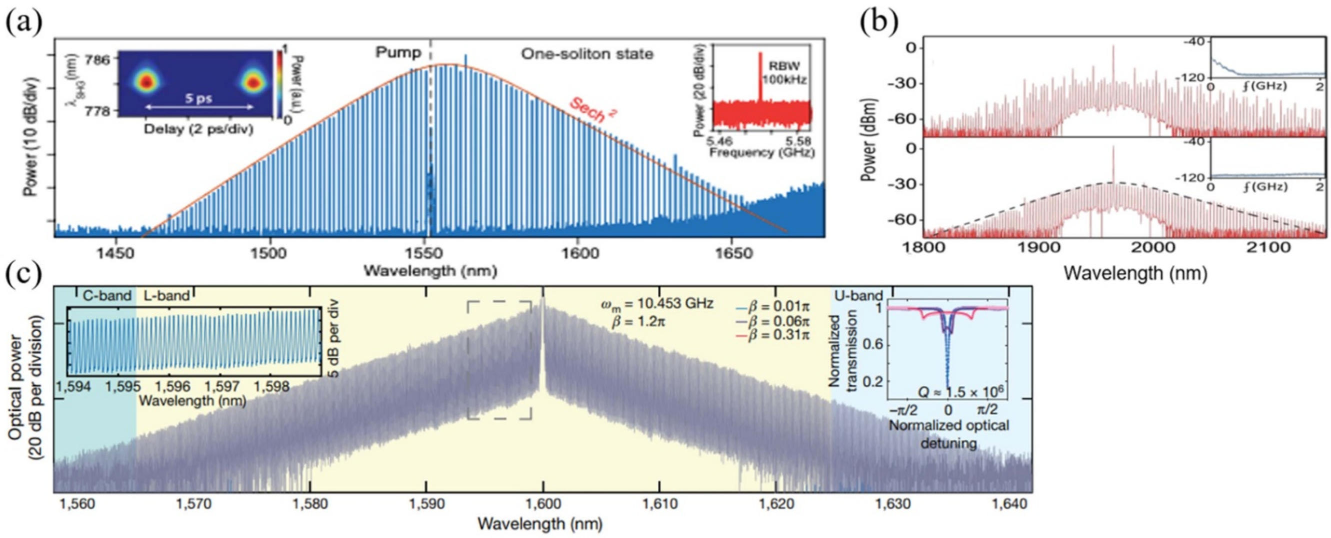

Figure 15. OFC based on LNOI platforms. (a) Soliton comb on Z-cut LN at 1.5 μm (Reproduced from [147]. CC BY 4.0). (b) Soliton Kerr combs on Z-cut LN generated at 2 μm in order to mitigate strong Raman gain (Reprinted with permission from [148] © The Optical Society). (c) Measured output spectrum of an EO frequency comb based on the LN microring resonator (Reproduced from [47], with permission from Springer Nature).

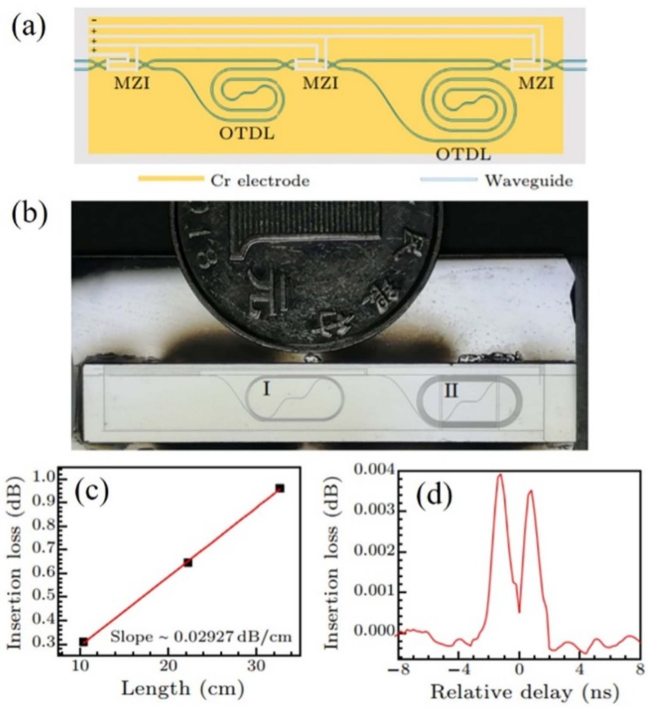

Figure 16. ODLs based on LNOI. (a) Schematic of MZI-switchable ODLs. (b) Micrograph of ODL device. (c) The measured losses of ODLs. (d) Time delay when the signal transmits from the shortest waveguide and two cascaded optical true delay lines (OTDLs) (Reproduced from [40]. © IOP Publishing Ltd. All rights reserved).

Based on the thin-film LN platform, EO-switchable ODLs were successfully realized based on LN ridge waveguides, and consisted of of two ODLs and three EO MZI switches (figure 16) [40]. As total length of these ODLs was over 1 m, one could obtain a delay time ~2.28 ns with a relatively low propagation loss ~0.029 dB cm^{-1}.

It is highly desirable to develop an on-chip spectrometer. Benefitting from the thin-film LN platform, an integrated single-waveguide Fourier transform spectrometer was demonstrated with an operation bandwidth of 500 nm in the near-infrared range (figure 17) [50]. The prototype device did not rely on any moving components and was based on an EO-induced optical delay, while preserving a footprint of less than 10 mm^2 [50].

4.6. Quantum photonics

Due to the large second-order nonlinearity of LN, spontaneous parametric down conversion (SPDC) has been widely investigated based on LNOI to create various applications in quantum optics, such as entangled photon pair generation and quantum memory. For an entangled photon pair source, it has been applied in quantum communication, computation, and metrology [153]. Tanzilli *et al* demonstrated energy–time and time-bin entangled photon pairs generated in a PPLN waveguide via SPDC, with a conversion efficiency up to 10^{-6} [154]. Another report on photon pairs generation based on a QPM PPLN ridge waveguide showed full width at half maximum of 1 nm and brightness of ~6 × 10^{-5} s GHz^{-1} [155]. Therefore, on-chip engineering of quantum light sources can be achieved by combining LN waveguide circuits and PPLN on a photonic integrated chip [25]. For quantum memory, erbium- and thulium-doped LN waveguides can be used to store quantum optic information for nanosecond and microsecond timescales [156]. Additionally, Zhang *et al* reported a new physical mechanism via two coupled LNOI-based EO microrings where the storage and retrieval of light were based on a photonic dark mode [48]. Most recently, Xu

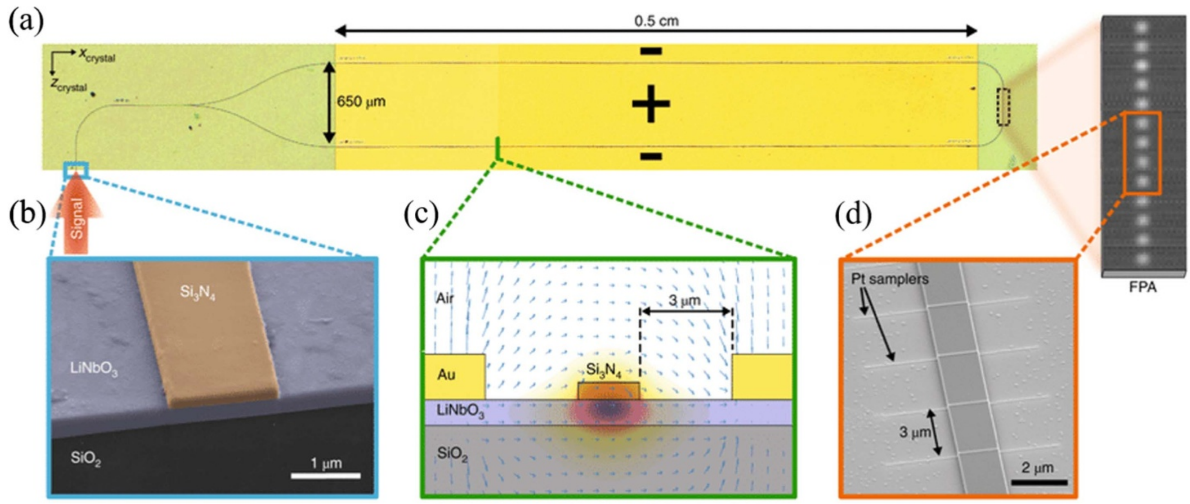

Figure 17. Electro-optic LN waveguide spectrometer. (a) Optical microscope image of the closed-loop LN–SiN waveguide structure with gold electrodes. (b) SEM image of the input of a SiN–LN ridge waveguide. (c) Cross-sectional schematic of the waveguide–electrode configuration. (d) SEM image of four platinum evanescent field samplers on top of the waveguide (Reproduced from [50], with permission from Springer Nature).

et al demonstrated a bright and broadband biphoton quantum source in an LN microdisk by natural phase matching in SPDC, with high purity and excellent interference visibility (96.5% $\pm$ 2%) [157].

5. Conclusion and future prospects

In conclusion, we have reviewed the recent progress regarding LN waveguide-based nonlinear photonic devices. Various aspects of this field have been introduced, including basic waveguide theory, fabrication techniques, diverse waveguide-based on-chip devices including wavelength converters, EO modulators, SCG, OFCs, ODLs, and on-chip spectrometers. It is important to mention that LN waveguide-based nonlinear photonic devices have been shown to be superior over traditional techniques in many aspects. However, it is still desirable to further optimize the fabrication techniques and improve their CMOS-compatibility, so as to transit from proof-of-concept demonstrations to scalable, low-cost, and high-yield enabling technologies and products. Furthermore, it has been suggested that the functionality of LN waveguide-based devices can be extended to quantum photonic circuits in the future, including quantum light sources, processors, and detectors. With these efforts, we believe that LN-based nonlinear photonic devices will make a profound impact on modern photonic society in the future.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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